Zdravko Botev
Alexander Keller
Christiane Lemieux
Bruno Tuffin *Editors*

# Advances in Modeling and Simulation

## Festschrift for Pierre L'Ecuyer

Springer

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# Preface

This Festschrift is a collection of invited research articles on the occasion of Pierre L'Ecuyer's 70th birthday in 2020. During the pandemic, a celebration with friends and colleagues was impossible. When discussing the opportunity to publish a Festschrift instead, support has been enthusiastic. We are grateful to the authors of this volume for their endorsement and their ready willingness to contribute. The works reflect Pierre's influence on the fields of stochastic modeling, simulation, and operations research. It is a real pleasure to present this Festschrift to honor Pierre L'Ecuyer.

Sydney, Australia — Zdravko Botev
Berlin, Germany — Alexander Keller
Waterloo, Canada — Christiane Lemieux
Rennes, France — Bruno Tuffin
April 2022

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Zdravko Botev
Alexander Keller
Christiane Lemieux
Bruno Tuffin

# Biography

**Abstract** Pierre L'Ecuyer is regarded as a top scientist, his leadership in the field of simulation and, in particular, pseudo-random number generation being uncontested. Pierre is not only known for his scientific results, but also for his rigorousness, his dedication to excellence, his dynamism, his enormous working capacity, and his curiosity making an impression on every person who gets to know him. Those in desire of more detail may have a look at Pierre's vita at http://www.iro.umontreal.ca/lecuyer/cva.pdf.

## Education

Pierre followed an education path from the Université de Montréal, Canada. His first academic degree is a Bachelor's in mathematics in 1972, followed by an M.Sc. in Operations Research in 1980 (in between he had been a teacher of mathematics at the CEGEP of Sept-Iles, Québec), and then a Ph.D. in Computer Science with Operations Research orientation in 1983 on Markovian decision processes. From 1982 to 1990, he served as a Professor in the Computer Science Department at Université Laval, and since then has been a Professor in the "Département d'informatique et de recherche opérationnelle" (DIRO) at the Université de Montréal. Pierre has also been spending time visiting colleagues in numerous places worldwide during sabbaticals or long stays, including Stanford University, Université de Nantes, Waseda University, University of Salzburg, North Carolina State University, Université de Savoie, Inria Rocquencourt and Rennes, University of New South Wales, and Google Research.

## Research Activity and Visibility

As of the beginning of November 2021, Pierre has written or co-written 26 book chapters, 125 journal papers (in prestigious journals), and 134 referred conference papers. According to Google Scholar, his H-index is 68 and he has 16,481 citations. Pierre has a wide range of interests and diverse contributions. Among these, Pierre is a world leader in multiple areas:

- **Dynamic programming and operations research (OR) in general**. Pierre's activity on dynamic programming (starting with his Ph.D. thesis) and a broad range of OR domains, including the computation of derivatives, is still highly referenced in the domain.
- **Variance reduction techniques**. His work on variance reduction, including rare event simulation, has made Pierre one of the most renowned researchers in the domain, both from the theoretical and application (to reliability and queuing) aspects. He contributed to the advances of importance sampling and splitting procedures drastically reducing the simulation time to reach a predefined accuracy.
- **Telephone call centers**. His industrial contracts have led Pierre to work on the modeling, simulation, and optimization of call centers. His activity has led to the development of a software used by several companies and the development of models and specific analysis techniques making him a renowned expert in the area collaborating with the best-known other teams in the world. He notably developed new and more realistic models than those existing at the time in which the arrival rate of customers changes with time, is stochastic, and the arrival rates in different time periods are not independent. Novel estimations of parameters have also been designed. He also developed simulation-based optimization algorithms and heuristics for agent's staffing.
- **Quasi-Monte Carlo methods** are deterministic methods, as opposed to the random Monte Carlo ones, having the advantage of converging faster, even if less easy to apply. Again, Pierre has developed a strong and world-leading expertise in the generation of sequences of highly uniformly distributed points used by those methods, their randomization to get a practical estimation of the error, and their application in finance.
- **Random number generators**. While Pierre's work on all the previously described topics is already impressive, his activity on random number generators, a key issue for simulation, is probably what should be highlighted the most. If one has to give a single name on this topic, Pierre is probably the one that will be mentioned. His random generator RNGStreams is widely used, because it is one of the most efficient and portable ones. His extensive test suites are also very popular. Pierre's TestU01 software library is the standard suite of procedures for empirically testing the performance of random number generators. Among his many publications on the subject, the one he has in the *Communications of the ACM* has been cited more than 1,100 times.

## Editorial Activities

Pierre is currently an Associate Editor for three journals: *ACM Transactions on Mathematical Software, Statistics and Computing*, and *International Transactions in Operational Research.* He was previously Associate Editor for five other journals and the Departmental Editor for the Simulation Department of Management Science. He was the Editor-in-Chief for *ACM Transactions on Modeling and Computer Simulation* from 2010 to 2013, a period during which the journal grew in scope and volume of submissions; Pierre was carefully reading all the submitted papers and prescreening submissions to alleviate the workload of the editorial team.

Remarkably, Pierre has reviewed articles for 170 *different* journals. That illustrates how well-known he is by people even outside the simulation community. Typically, he has been reviewing between three to four papers *per week* on average.

## Organizing Conferences

Pierre has already organized seven international events in Montréal, including the Eighth International Conference on Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing (MCQMC) in 2008 and the INFORMS Simulation Society Workshop in 2011, or the Eleventh International Conference on Monte Carlo Methods and Application (MCM) in 2017. He also co-organized MCQMC 2018 in Rennes, France. Pierre has been serving on many program or steering committees.

## Simulation Societies

Pierre has been a member of numerous evaluation committees worldwide for grant proposals, promotion, and prizes from universities. Among the most notable memberships are the INFORMS Simulation Society Distinguished Service Award, INFORMS Simulation Society Outstanding Publication Award, or INFORMS College on Simulation Outstanding Publication and Outstanding Award committees.

## Industry and Civil Society

Pierre has received many grants from the industry to successfully apply his simulation and operations research results. For example, Pierre has developed specific modeling and simulation tools for Bell and Hydro-Québec. He has also been contacted to implement his random number generators, or test existing ones, by AMD, Alcatel, LottoQuébec, The Mathworks, and Montréal Police Service, to name a few.

Pierre has devoted a lot of time to software development, with all his random generators available in many languages, a Java library for stochastic simulation called SSJ, or a software library called TestU01 offering a collection of utilities for the empirical statistical testing of uniform random number generators. All these tools are freely available to the scientific community.

## Mentorship

Pierre has supervised 49 Master and Ph.D. students, as well as 23 postdocs. He has been a member of numerous Ph.D. examination committees all over the world. His courses on simulation at the Université de Montréal and various Summer Schools have also contributed to the widespread dissemination of modeling and simulation knowledge.

## Recognition

Pierre has been recognized by the scientific community, having received prestigious awards such as a Canadian and an Inria Research Chair, the Award of Merit from the Canadian Operational Research Society, the INFORMS Fellow Award in 2006, the INFORMS Simulation Society Distinguished Service Award in 2011, the INFORMS Simulation Society Outstanding Research Publication Award won three times in 1999 (on Combined Multiple Recursive Random Number Generators), 2009 (on computational finance, by designing efficient algorithms for pricing path-dependent options), and 2018 (on call centers modeling), the SIGSIM Distinguished Contributions Award in 2016, or INFORMS Simulation Society Lifetime Professional Achievement Award in 2020. On the Canadian scene, to name only a few of the awards and distinctions earned by Pierre, in 1996, he was awarded a prestigious Steacie Fellowship for the period 1995–1997 from the Natural Science and Engineering Research Council of Canada. He received a Killam Research Fellowship from the Canada Council for the Arts for the period 2001–2003. And in 2004, he was awarded a Canada Research Chair on Stochastic Simulation and Optimization for the period 2004–2010, which was renewed in 2011 for another seven years.

## Personal Achievements

Besides his scientific life, Pierre has had an amazing sport-related life, for which he is also well-known in the respective communities: not only as a competitor but also as a coach. While skilled in many kinds of sport, Pierre is exceptionally competitive and skilled in cross-country skiing and road cycling.

In cross-country skiing, he won the bronze medal at the Canadian championship in 50 km classic in 1994. But his main achievements are probably in road cycling. He was Canadian champion (by age groups) in 2000, 2001, 2011, and 2012 and finished second in 2002 and 2004. He was Quebec champion in road race in 1996, 2000, 2001, 2003, and 2012; in time trial in 2002, 2003, 2004, 2005, and 2012; and in criterium in 2011 and 2016. Among successful participation in other races, he won the America's cup in 2000 and 2012. He was named cyclist of the year 2012 in masters categories by the Quebec Cycling Federation, another award in another category! Pierre keeps on riding his bicycle all over the world and is known to always bring his bike with him when traveling.

Prior to these accomplishments, Pierre had been a coach in Track and Field between 1970 and 1992. He started by building and training a local team. He became a member of the Canadian team at the Olympic Games, World Championships, Commonwealth Games, etc. He was named Quebec's track and field federation's "coach of the year" in 1985 and 1992. He was the coach of the Barcelona Olympic games silver medalist (20 km walk) and world record holder (30 km walk) Guillaume Leblanc, from 1973 to 1992.

# Contents

**Monte Carlo Methods for Pricing American Options** . . . . . 1
Raul Chavez Aquino, Fabian Bastin, Maria Benazzouz, and Mohamed Kharrat

**Remarks on Lévy Process Simulation** . . . . . 21
Søren Asmussen

**Exact Sampling for the Maximum of Infinite Memory Gaussian Processes** . . . . . 41
Jose Blanchet, Lin Chen, and Jing Dong

**Truncated Multivariate Student Computations via Exponential Tilting** . . . . . 65
Zdravko I. Botev and Yi-Lung Chen

**Quasi-Monte Carlo Methods in Portfolio Selection with Many Constraints** . . . . . 89
Alexander Brunhuemer and Gerhard Larcher

**Geometric-Moment Contraction of G/G/1 Waiting Times** . . . . . 111
Kemal Dinçer Dingeç, Christos Alexopoulos, David Goldsman, Athanasios Lolos, and James R. Wilson

**Tractability of Approximation in the Weighted Korobov Space in the Worst-Case Setting** . . . . . 131
Adrian Ebert, Peter Kritzer, and Friedrich Pillichshammer

**Rare-Event Simulation via Neural Networks** . . . . . 151
Lachlan J. Gibson and Dirk P. Kroese

**Preintegration is Not Smoothing When Monotonicity Fails** . . . . . 169
Alexander D. Gilbert, Frances Y. Kuo, and Ian H. Sloan

**Combined Derivative Estimators** . . . . . 193
Paul Glasserman

**A Central Limit Theorem For Empirical Quantiles in the Markov Chain Setting** . . . . . 211
Peter W. Glynn and Shane G. Henderson

**Simulation of Markov Chains with Continuous State Space by Using Simple Stratified and Sudoku Latin Square Sampling** . . . . . 239
Rami El Haddad, Joseph El Maalouf, Rana Fakhereddine, and Christian Lécot

**Quasi-Random Sampling with Black Box or Acceptance-Rejection Inputs** . . . . . 261
Erik Hintz, Marius Hofert, and Christiane Lemieux

**A Generalized Transformed Density Rejection Algorithm** . . . . . 283
Wolfgang Hörmann and Josef Leydold

**Fast Automatic Bayesian Cubature Using Sobol' Sampling** . . . . . 301
Rathinavel Jagadeeswaran and Fred J. Hickernell

**Rendering Along the Hilbert Curve** . . . . . 319
Alexander Keller, Carsten Wächter, and Nikolaus Binder

**Array-RQMC to Speed up the Simulation for Estimating the Hitting-Time Distribution to a Rare Set of a Regenerative System** . . . . . 333
Marvin K. Nakayama and Bruno Tuffin

**Foundations of Ranking & Selection for Simulation Optimization** . . . . . 353
Barry L. Nelson

**Where are the Logs?** . . . . . 381
Art B. Owen and Zexin Pan

**Network Reliability, Performability Metrics, Rare Events and Standard Monte Carlo** . . . . . 401
Gerardo Rubino

# Fast Automatic Bayesian Cubature Using Sobol' Sampling




**Rathinavel Jagadeeswaran and Fred J. Hickernell**

**Abstract** Automatic cubatures approximate integrals to user-specified error tolerances. For high dimensional problems, it is difficult to adaptively change the sampling pattern to focus on peaks because peaks can hide more easily in high dimensional space. But, one can automatically determine the sample size, $n$, given a reasonable, fixed sampling pattern. This approach is pursued in Jagadeeswaran and Hickernell, Stat. Comput., 29:1214–1229, 2019, where a Bayesian perspective is used to construct a credible interval for the integral, and the computation is terminated when the half-width of the interval is no greater than the required error tolerance. Our earlier work employs integration lattice sampling, and the computations are expedited by the fast Fourier transform because the covariance kernels for the Gaussian process prior on the integrand are chosen to be shift-invariant. In this chapter, we extend our fast automatic Bayesian cubature to digital net sampling via *digitally* shift-invariant covariance kernels and fast Walsh transforms. Our algorithm is implemented in the MATLAB Guaranteed Automatic Integration Library (GAIL) and the QMCPy Python library.

**Keywords** Adaptive multivariate cubature · Probabilistic numerics · Digital nets · Stopping criteria · GAIL · QMCPy

## 1 Introduction

Cubature, or numerical multivariate integration, is the problem of inferring a numerical value for a definite integral,

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$$\mu := \mu(f) := \int_{[0,1]^d} f(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x}, \tag{1}$$

when no closed-form analytic expression exists or is easily available. Typically, values of $f$ are accessible through a black-box function routine. Our goal is to construct a cubature, $\widehat{\mu}_n = \widehat{\mu}_n(f)$, depending only on integrand values at the nodes $\{\boldsymbol{x}_i\}_{i=1}^n$, and determine the $n$ that satisfies the error criterion

$$|\mu - \widehat{\mu}_n| \le \varepsilon \tag{2}$$

with high probability. This article extends the fast Bayesian cubature ideas presented in [12] to digital sequences [8].

Cubature is a key component of many problems in scientific computing, finance [10], statistical modeling, imaging [14], uncertainty quantification, and machine learning [11]. The original form of the integral may require a suitable variable transformation to become (1). This process is addressed in [1, 6, 15, 22, 23].

Following the Bayesian numerics approach of [3, 7, 19, 20] and others, we assume that our integrand, $f$, is an instance of a Gaussian process, $\mathcal{GP}(m, s^2 C_{\boldsymbol{\theta}})$, and construct a probabilistic error bound for $\mu$ via a Bayesian credible interval. Here, the random function $f$ has constant mean, $m$, and covariance kernel $s^2 C_{\boldsymbol{\theta}}$, where $s$ is a positive scale factor, and $C_{\boldsymbol{\theta}} : [0,1]^d \times [0,1]^d \to \mathbb{R}$ is a symmetric, positive-definite kernel parameterized by $\boldsymbol{\theta}$. The parameter $\boldsymbol{\theta}$ may affect the shape or smoothness of $C_{\boldsymbol{\theta}}$. The integrand is sampled until the credible interval becomes small enough to satisfy (2) with high probability.

Our approach to *fast* Bayesian cubature [12] relies on two key points:

(i) Choosing covariance kernels, $C_{\boldsymbol{\theta}} : [0,1]^d \times [0,1]^d \to \mathbb{R}$, for which the symmetric, positive definite Gram matrices,

$$\mathsf{C}_{\boldsymbol{\theta}} = \big(C_{\boldsymbol{\theta}}(\boldsymbol{x}_i, \boldsymbol{x}_j)\big)_{i,j=1}^n, = (\boldsymbol{C}_{\boldsymbol{\theta},1}, \ldots, \boldsymbol{C}_{\boldsymbol{\theta},n}), \tag{3}$$

have an eigenvalue-eigenvector decomposition of the form[1] $\mathsf{C}_{\boldsymbol{\theta}} = \mathsf{V}\Lambda_{\boldsymbol{\theta}}\mathsf{V}^H/n$, where

$$\mathsf{V} \text{ may be identified analytically,} \tag{4a}$$

$$\text{The first row and column of } \mathsf{V} \text{ are } \mathbf{1}, \tag{4b}$$

$$\text{Computing } \mathsf{V}^H \boldsymbol{b} \text{ requires only } \mathcal{O}(n \log n) \text{ operations } \forall \boldsymbol{b}, \text{ and} \tag{4c}$$

$$\int_{[0,1]^d} C_{\boldsymbol{\theta}}(\boldsymbol{t}, \boldsymbol{x}) \, \mathrm{d}\boldsymbol{t} = 1 \qquad \forall \boldsymbol{x} \in [0,1]^{\mathrm{d}}, \tag{4d}$$

and

[1] The presence of $1/n$ in the eigenvalue-eigenvector decomposition arises from the assumption that the first column of $\mathsf{V}$ is $\mathbf{1}$. It could be removed by assuming that the first column of $\mathsf{V}$ is $\mathbf{1}/\sqrt{n}$. The superscript $H$ denotes the complex conjugate transpose [2].

(ii) The hyperparameters of the Gaussian process—$m$, $s$, and $\boldsymbol{\theta}$—are tuned to increase the likelihood that $f$ is a typical integrand and not an outlier.

We call the transformation $\boldsymbol{b} \mapsto \mathsf{V}^H \boldsymbol{b}$ a *fast Bayesian transform*. Our earlier work [12] focuses on lattice nodes and shift-invariant kernels. In that context the computation of $\mathsf{V}^H \boldsymbol{b}$ is a one-dimensional fast Fourier transform. This chapter focuses on digital sequences, such as Sobol' sequences [24], and covariance kernels that are digitally shift-invariant. In this case the computation of $\mathsf{V}^H \boldsymbol{b}$ is a fast Walsh-Hadamard transform.

The next section summarizes the key formulae from [12]. Section 3 extends fast Bayesian cubature to digital nets and digital-shift invariant kernels defined in terms of Walsh functions. Section 4 presents numerical experiments that illustrate these ideas.

## 2 Bayesian Cubature

As noted above, we assume the integrand, $f$, is an instance of a real-valued stochastic Gaussian process, i.e., $f \sim \mathcal{GP}(m, s^2 C_{\boldsymbol{\theta}})$. Let $\boldsymbol{y} = (y_i)_{i=1}^n = \big(f(\boldsymbol{x}_i)\big)_{i=1}^n$ denote the vector of sampled integrand values. In [12] we introduce three methods for resolving the hyperparameters: empirical Bayes (EB), full Bayes (full), and generalized cross-validation (GCV). Under assumptions (4a)–(4d) for the covariance kernel and sampling sites, it is shown in [12, Theorem 2] that the credible interval for the integral takes the form

$$\mathbb{P}_f \left[ |\mu - \widehat{\mu}_n| \le \mathrm{err}_{\mathrm{CI}} \right] = 99\%,$$

where

$$\begin{aligned}
\widehat{\mu}_n &= \frac{1}{n} \sum_{i=1}^n y_i = \widetilde{y}_1, \quad \text{where } y_i = f(\boldsymbol{x}_i), \qquad (5)\\
\widetilde{\boldsymbol{y}} &:= \mathsf{V}^H \boldsymbol{y}, \quad \text{where } \widetilde{\boldsymbol{y}} \text{ is the fast transform of } \boldsymbol{y},\\
\boldsymbol{\lambda}_{\boldsymbol{\theta}} &:= \begin{pmatrix} \lambda_{\boldsymbol{\theta},1} \\ \vdots \\ \lambda_{\boldsymbol{\theta},n} \end{pmatrix} = \mathsf{V}^H \boldsymbol{C}_{\boldsymbol{\theta},1},\\
&\quad \text{where } \boldsymbol{C}_{\boldsymbol{\theta},1} \text{ is the first column of the Gram matrix (3)},\\
s^2_{\mathrm{EB}} &= \frac{1}{n^2} \sum_{i=2}^n \frac{|\widetilde{y}_i^2|}{\lambda_{\boldsymbol{\theta},i}},\\
s^2_{\mathrm{GCV}} &= \frac{1}{n} \sum_{i=2}^n \frac{|\widetilde{y}_i^2|}{\lambda^2_{\boldsymbol{\theta},i}} \left[ \sum_{i=1}^n \frac{1}{\lambda_{\boldsymbol{\theta},i}} \right]^{-1},
\end{aligned}$$

$$\widehat{\sigma}^2_{\text{full}} = \frac{1}{n(n-1)} \sum_{i=2}^{n} \frac{|\widetilde{y}_i^2|}{\lambda_{\boldsymbol{\theta},i}} \left( \frac{\lambda_{\boldsymbol{\theta},1}}{n} - 1 \right),$$

$$\boldsymbol{\theta}_{\text{EB}} = \operatorname*{argmin}_{\boldsymbol{\theta}} \left[ \log \left( \sum_{i=2}^{n} \frac{|\widetilde{y}_i^2|}{\lambda_{\boldsymbol{\theta},i}} \right) + \frac{1}{n} \sum_{i=1}^{n} \log(\lambda_{\boldsymbol{\theta},i}) \right], \tag{6a}$$

$$\boldsymbol{\theta}_{\text{GCV}} = \operatorname*{argmin}_{\boldsymbol{\theta}} \left[ \log \left( \sum_{i=2}^{n} \frac{|\widetilde{y}_i^2|}{\lambda_{\boldsymbol{\theta},i}^2} \right) - 2 \log \left( \sum_{i=1}^{n} \frac{1}{\lambda_{\boldsymbol{\theta},i}} \right) \right]. \tag{6b}$$

$$\text{err}_{\text{CI},\mathsf{x}} = \text{err}_{\mathsf{x}} = 2.58 s_{\mathsf{x}} \sqrt{1 - \frac{n}{\lambda_{\boldsymbol{\theta}_{\mathsf{x}},1}}}, \qquad \mathsf{x} \in \{\text{EB}, \text{GCV}\}, \tag{7a}$$

$$\text{err}_{\text{CI,full}} = t_{n-1,0.995} \widehat{\sigma}_{\text{full}} > \text{err}_{\text{EB}}. \tag{7b}$$

In the formulas for the credible interval half-widths, $\boldsymbol{\theta}$ is assumed to take on the values $\boldsymbol{\theta}_{\text{EB}}$ or $\boldsymbol{\theta}_{\text{GCV}}$ as appropriate. There is no suitable $\boldsymbol{\theta}_{\text{full}}$.

Our Bayesian cubature algorithm doubles the sample size until the width of the credible interval is small enough. Doubling the sample size allows us to retain the preferred structure of the sample sites. The Bayesian cubature steps are detailed in Algorithm 1.

**Algorithm 1:** Automatic Bayesian Cubature

**Input**: a generator for the sequence $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots$; a black-box function, $f$; an absolute error tolerance, $\varepsilon > 0$; the positive initial sample size, $n_0$; the maximum sample size $n_{\max}$

$n \leftarrow n_0$, $n' \leftarrow 0$, $\text{err}_{\text{CI}} \leftarrow \infty$;
**while** $\text{err}_{\text{CI}} > \varepsilon$ *and* $n \le n_{\max}$ **do**
  Generate $\{\boldsymbol{x}_i\}_{i=n'+1}^{n}$ and sample $\{f(\boldsymbol{x}_i)\}_{i=n'+1}^{n}$;
  Compute $\boldsymbol{\theta}$ by (6a) or (6b);
  Compute $\text{err}_{\text{CI}}$ according to (7a) or (7b);
  $n' \leftarrow n$, $n \leftarrow 2n'$;
Sample size to compute $\widehat{\mu}_n$, $n \leftarrow n'$;
Compute $\widehat{\mu}_n$, the approximate integral, according to (5);
**return** $\widehat{\mu}_n$, *n, and* $\text{err}_{\text{CI}}$;

We recognize that multiple applications of our credible intervals in one run of the algorithm is not strictly justified. However, if our integrand comes from the middle of the sample space and not the extremes, we expect our automatic Bayesian cubature to approximate the integral within the desired error tolerance with high probability. The examples in Sect. 4 support that expectation.

The credible intervals in our automatic algorithm are homogeneous with respect to the function data. If they are valid for some integrand, $f$, they are also valid for the integrand $af$ for any constant $a$.

## 3 Digital Nets and Walsh Kernels

The previous section does not mention which sequences of data sites and kernels satisfy assumptions (4a, 4b, 4c, 4d). We demonstrate in [12] that rank-1 lattice points and shift-invariant kernels do so. In this section, we give another example, namely digital sequences and digitally shift-invariant kernels based on Walsh functions. For completeness, we define digital sequences, digitally shift-invariant kernels, and the fast Walsh-Hadamard transform.

### *3.1 Digital Sequences*

The first example of a digital sequence was proposed by Sobol' [24], and it is also the most popular digital sequence. This chapter focuses on digital sequences in base 2, which includes Sobol' sequences.

**Definition 1** Let $(\,)_2$ denote the base 2 or binary expansion of a number. For any non-negative integer $i = (\ldots i_3 i_2 i_1)_2$, define $\overrightarrow{i} = (i_1, i_2, \ldots)^T$ as the $\infty \times 1$ vector $\overrightarrow{i}$ of its binary digits. For any point $z = (0.z_1 z_2 \ldots)_2 \in [0, 1)$, define $\overrightarrow{z} = (z_1, z_2, \ldots)^T$ as the $\infty \times 1$ vector of its binary digits. Let $\mathsf{G}_1, \ldots, \mathsf{G}_d$ denote predetermined $\infty \times \infty$ generator matrices whose elements are zeros and ones. A *digital sequence* in base 2 is $\{\boldsymbol{z}_1, \boldsymbol{z}_2, \boldsymbol{z}_3, \ldots\}$, where each $\boldsymbol{z}_i = (z_{i1}, \ldots, z_{id})^T \in [0, 1)^d$ is defined by

$$\overrightarrow{z}_{i+1,\ell} = \mathsf{G}_\ell \overrightarrow{i} \pmod 2, \quad \ell = 1, \ldots, d, \quad i = 0, 1, \ldots .$$

For any non-negative integers $m$ and $\tau$, the set $\{\boldsymbol{z}_{\tau 2^m+1}, \ldots, \boldsymbol{z}_{(\tau+1)2^m+1}\}$ is a *digital net* in base 2.

It is common to index digital sequences starting with 0, whereas our data sites and matrices are indexed starting with 1. To keep our notation consistent, we define $\boldsymbol{z}_{i+1}$ in terms of $\overrightarrow{i}$.

Digital sequences have a group structure under digit-wise, element-by-element addition modulo the base, which we denote by $\oplus$ and which also corresponds to an exclusive-or in base 2. Here and in what follows we ignore the cases of measure zero for which the $\oplus$ operation leads to a binary representation ending in an infinite string of ones. The following lemma summarizes some important properties of digital sequences.

**Lemma 1** *Let $\{z_i\}_{i=1}^{\infty}$ be a digital sequence in base 2 as defined in Definition 1. Choose any digital shift $\boldsymbol{\Delta} \in [0,1)^d$, and define the sequence of nodes $\{\boldsymbol{x}_i\}_{i=1}^{\infty}$ by digitwise addition, $\boldsymbol{x}_i = \boldsymbol{z}_i \oplus \boldsymbol{\Delta}$. Then for all $i, j \in \mathbb{N}_0$,*

$$\boldsymbol{x}_{i+1} \oplus \boldsymbol{x}_{i+1} = \boldsymbol{0}, \quad \boldsymbol{x}_{i+1} \oplus \boldsymbol{x}_{j+1} = \boldsymbol{x}_{j+1} \oplus \boldsymbol{x}_{i+1} = \boldsymbol{z}_{i+1} \oplus \boldsymbol{z}_{j+1} = \boldsymbol{z}_{(i \oplus j)+1}.$$

*Therefore, $\{\boldsymbol{z}_i\}_{i=1}^{\infty}$ is a group. Moreover, the digital net $\{\boldsymbol{z}_i\}_{i=1}^{2^m}$ is a subgroup for $m \in \mathbb{N}_0$.*

The proof follows from Definition 1, and can be found in [21].

Digital sequence generators can be chosen by number theory, as are those of Sobol' [24] and Niederreiter and Xing [17] (see also [8, Chap. 8]), or they can be chosen by computer search [8, Chap. 10]. The original generator matrices may be scrambled using linear matrix scrambling [16].

## 3.2 Covariance Kernels Constructed Via Walsh Functions

The digitally shift invariant kernels required for fast Bayesian cubature using digital nets are constructed via Walsh functions, again specializing to base 2. The one-dimensional Walsh functions in base 2 are defined as

$$\begin{aligned} \mathrm{wal}_k(x) &:= (-1)^{k_0 x_1 + k_1 x_2 + \cdots} = (-1)^{\langle \vec{k}, \vec{x} \rangle}, \qquad k \in \mathbb{N}_0, \ x \in [0,1), \\ \langle \vec{k}, \vec{x} \rangle &:= k_0 x_1 + k_1 x_2 + \cdots . \end{aligned}$$

where again $\vec{k}$ is a vector containing the binary digits of $k$, and $\vec{x}$ is a vector containing the binary digits of $x$. Note that by this definition, $\mathrm{wal}_k(x \oplus t) = \mathrm{wal}_k(x)\mathrm{wal}_k(t)$.

These Walsh functions can be used to construct a covariance kernel for univariate integrands as follows

$$\begin{aligned} C_{\boldsymbol{\theta}}(x,t) &= K_{\boldsymbol{\theta}}(x \oplus t), \qquad K_{\boldsymbol{\theta}}(x) := 1 + \eta \omega_r(x), \qquad \boldsymbol{\theta} = (r, \eta), \\ \omega_r(x) &:= \sum_{k=1}^{\infty} \frac{\mathrm{wal}_k(x)}{2^{2r\lfloor \log_2 k \rfloor}}, \qquad r \in \mathbb{N}, \qquad \eta \in (0,1) \\ \omega_r(x_\ell \oplus t_\ell) &= \sum_{k=1}^{\infty} \frac{\mathrm{wal}_k(x)\mathrm{wal}_k(t)}{2^{2r\lfloor \log_2 k \rfloor}}. \end{aligned}$$

The symmetric, positive definite property of $C_{\boldsymbol{\theta}}(x,t)$ follows from its definition. This kernel is digitally shift invariant because

$$C_{\boldsymbol{\theta}}(x \oplus \Delta, t \oplus \Delta) = K_{\boldsymbol{\theta}}(x \oplus \Delta \oplus t \oplus \Delta) = K_{\boldsymbol{\theta}}(x \oplus t) = C_{\boldsymbol{\theta}}(x,t).$$

This follows because $\Delta \oplus \Delta = 0$ for any $\Delta \in [0,1)$.

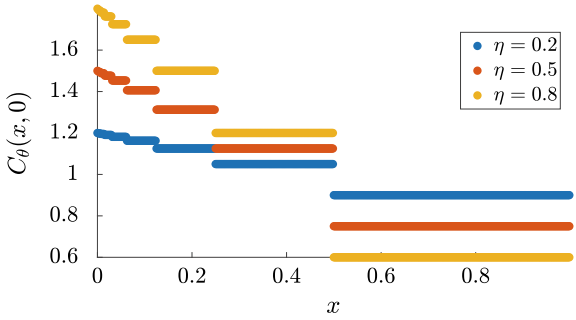



**Fig. 1** Walsh kernel of order $r = 1$ in dimension $d = 1$. This figure can be reproduced using `plot_walsh_kernel.m`

The parameter $r$ is a measure of the digital smoothness of the kernel, which does not correspond to ordinary smoothness. An explicit expression is available for $\omega_r$ in the case of order $r = 1$ (see [18]):

$$\omega_1(x) = 1 - 6 \times 2^{\lfloor \log_2 x \rfloor - 1}.$$

Figure 1 shows $C_{\boldsymbol{\theta}}(x, t)$ for order $r = 1$ and various values of $\eta$ in the interval $[0, 1)$. Unlike the shift-invariant kernels used with lattice nodes, Walsh kernels are discontinuous and piecewise constant.

Covariance kernels for multivariate integrands defined on $[0, 1)^d$ are constructed as tensor products:

$$C_{\boldsymbol{\theta}}(\boldsymbol{x}, \boldsymbol{t}) = K_{\boldsymbol{\theta}}(\boldsymbol{x} \oplus \boldsymbol{t}), \tag{8}$$

$$K_{\boldsymbol{\theta}}(\boldsymbol{x}) = \prod_{\ell=1}^{d} [1 + \eta_\ell \omega_r(x_\ell)], \quad \boldsymbol{\eta} = (\eta_1, \ldots, \eta_d), \quad \boldsymbol{\theta} = (r, \boldsymbol{\eta}). \tag{9}$$

For multidimensional kernels, smaller $\eta_\ell$ for $\ell \in (1, \ldots, d)$ implies less variation in the amplitude of the kernel in dimension $\ell$. The parameter vector $\boldsymbol{\theta}$ now is of dimension $d + 1$. One might also choose $\eta_1 = \cdots = \eta_d = \eta$, in which case the parameter vector $\boldsymbol{\theta} = (r, \eta)$ has dimension two.

### 3.3 *Eigenvector-Eigenvalue Decomposition of the Gram Matrix*

For fast Bayesian cubature to succeed, the digital net data sites (Sect. 3.1) and the covariance kernels (Sect. 3.2) must match in a way to satisfy conditions (4a, 4b, 4c, 4d). To do this, we notice that the eigenvectors of the Gram matrix defined in (3) are the columns of the Walsh-Hadamard matrices, defined as follows:

$$\mathsf{H}^{0)} = 1, \quad \mathsf{H}^{(1)} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathsf{H}^{(2)} = \mathsf{H}^{(1)} \otimes \mathsf{H}^{(1)} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \ldots$$

$$\mathsf{H}^{(m)} = \mathsf{H}^{(m-1)} \otimes \mathsf{H}^{(1)} = \begin{pmatrix} \mathsf{H}^{(m-1)} & \mathsf{H}^{(m-1)} \\ \mathsf{H}^{(m-1)} & -\mathsf{H}^{(m-1)} \end{pmatrix} = \mathsf{H}^{(1)} \underbrace{\otimes \cdots \otimes}_{m \text{ times}} \mathsf{H}^{(1)}, \tag{10}$$

where $\otimes$ is Kronecker product. Note that the Walsh-Hadamard matrices are symmetric.

**Lemma 2** *Let $\{\boldsymbol{x}_i\}_{i=1}^{2^m}$ be nodes from a digitally shifted digital net in base 2, and let the covariance kernel take the form of (8). Then, for $m = 1, 2, \ldots$ the Gram matrix, $\mathsf{C}_{\boldsymbol{\theta}}^{(m)} = \big(C_{\boldsymbol{\theta}}(\boldsymbol{x}_i, \boldsymbol{x}_j)\big)_{i,j=1}^{2^m} = \big(K(\boldsymbol{x}_i \oplus \boldsymbol{x}_j)\big)_{i,j=1}^{2^m}$ is a $2 \times 2$ block-Hankel matrix [2, Definition 3.1.3], and all the sub-blocks and their sub-sub-blocks, etc. are also $2 \times 2$ block-Hankel. Moreover, $\mathsf{C}_{\boldsymbol{\theta}}^{(m)} \mathsf{H}^{(m)} = \mathsf{H}^{(m)} \Lambda_{\boldsymbol{\theta}}^{(m)}$, where $\Lambda_{\boldsymbol{\theta}}^{(m)}$ is the diagonal matrix of eigenvalues of $\mathsf{C}_{\boldsymbol{\theta}}^{(m)}$.*

***Proof.*** First define the following matrices using the notation from Lemma 1:

$$\begin{aligned} \mathsf{C}_{\boldsymbol{\theta}}^{(m,\tau)} &= \big(K(\boldsymbol{x}_i \oplus \boldsymbol{x}_{j+\tau 2^m})\big)_{i,j=1}^{2^m} \\ &= \big(K(\boldsymbol{z}_{i \oplus j+\tau 2^m+1})\big)_{i,j=0}^{2^m-1}, \qquad m, \tau = 0, 1, \ldots. \end{aligned}$$

This implies that $\mathsf{C}_{\boldsymbol{\theta}}^{(m+1,\tau)}$, has the following block structure:

$$\begin{aligned} &\mathsf{C}_{\boldsymbol{\theta}}^{(m+1,\tau)} \\ &\quad = \begin{pmatrix} \big(K(\boldsymbol{z}_{i \oplus j+\tau 2^{m+1}+1})\big)_{i,j=0}^{2^m-1} & \big(K(\boldsymbol{z}_{i \oplus (j+2^m)+\tau 2^{m+1}+1})\big)_{i,j=0}^{2^m-1} \\ \big(K(\boldsymbol{z}_{(i+2^m) \oplus j+\tau 2^{m+1}+1})\big)_{i,j=0}^{2^m-1} & \big(K(\boldsymbol{z}_{(i+2^m) \oplus (j+2^m)+\tau 2^{m+1}+1})\big)_{i,j=0}^{2^m-1} \end{pmatrix} \\ &\quad = \begin{pmatrix} \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)} & \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)} \\ \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)} & \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)} \end{pmatrix}, \end{aligned} \tag{11}$$

since for $i, j = 0, \ldots, 2^m - 1$, it follows that

$$\begin{aligned} i \oplus (j + 2^m) &= (i \oplus j) + 2^m = (i + 2^m) \oplus j, \\ (i + 2^m) \oplus (j + 2^m) &= (i \oplus j) + (2^m \oplus 2^m) = i \oplus j. \end{aligned}$$

The proof of (11) follows by induction. Note that for the case $m = 1$ and $\tau = 0, 1, \ldots$

$$\mathsf{C}_{\boldsymbol{\theta}}^{(1,\tau)} = \begin{pmatrix} \mathsf{C}_{\boldsymbol{\theta}}^{(0,2\tau)} & \mathsf{C}_{\boldsymbol{\theta}}^{(0,2\tau+1)} \\ \mathsf{C}_{\boldsymbol{\theta}}^{(0,2\tau+1)} & \mathsf{C}_{\boldsymbol{\theta}}^{(0,2\tau)} \end{pmatrix} = \begin{pmatrix} K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) & K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) \\ K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) & K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) \end{pmatrix},$$

which has the desired Hankel property. Note also the eigenvectors of $\mathsf{C}_{\boldsymbol{\theta}}^{(1,\tau)}$ are the columns of $\mathsf{H}^{(1)}$ since

$$\begin{aligned}\mathsf{C}_{\boldsymbol{\theta}}^{(1,\tau)}\mathsf{H}^{(1)} &= \begin{pmatrix} K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) + K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) & K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) - K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) \\ K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) + K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) & K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) - K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) \end{pmatrix} \\ &= \mathsf{H}^{(1)} \underbrace{\begin{pmatrix} K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) + K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) & 0 \\ 0 & K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+1}) - K_{\boldsymbol{\theta}}(\boldsymbol{z}_{2\tau+2}) \end{pmatrix}}_{\Lambda_{\boldsymbol{\theta}}^{(1,\tau)}}.\end{aligned}$$

Now assume that $\mathsf{C}_{\boldsymbol{\theta}}^{(m,\tau)}$ is $2 \times 2$ block-Hankel matrix with all its sub-blocks and sub-sub-blocks, etc. also $2 \times 2$ block-Hankel for $\tau = 0, 1, \ldots$. Then by (11), the same holds for $\mathsf{C}_{\boldsymbol{\theta}}^{(m+1,\tau)}$ for $\tau = 0, 1, \ldots$.

Moreover, the hypothesized eigenvectors of $\mathsf{C}_{\boldsymbol{\theta}}^{(m+1,\tau)}$ can also be verified by direct calculation under the induction hypothesis:

$$\begin{aligned}&\mathsf{C}_{\boldsymbol{\theta}}^{(m+1,\tau)}\mathsf{H}^{(m+1)} \\ &\quad= \begin{pmatrix} \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)} & \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)} \\ \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)} & \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)} \end{pmatrix} \begin{pmatrix} \mathsf{H}^{(m)} & \mathsf{H}^{(m)} \\ \mathsf{H}^{(m)} & -\mathsf{H}^{(m)} \end{pmatrix} \\ &\quad= \begin{pmatrix} [\mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)} + \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)}]\mathsf{H}^{(m)} & [\mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)} - \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)}]\mathsf{H}^{(m)} \\ [\mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)} + \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)}]\mathsf{H}^{(m)} & [\mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau+1)} - \mathsf{C}_{\boldsymbol{\theta}}^{(m,2\tau)}]\mathsf{H}^{(m)} \end{pmatrix} \\ &\quad= \begin{pmatrix} \mathsf{H}^{(m)} & \mathsf{H}^{(m)} \\ \mathsf{H}^{(m)} & -\mathsf{H}^{(m)} \end{pmatrix} \underbrace{\begin{pmatrix} \Lambda_{\boldsymbol{\theta}}^{(m,2\tau)} + \Lambda_{\boldsymbol{\theta}}^{(m,2\tau+1)} & 0 \\ 0 & \Lambda_{\boldsymbol{\theta}}^{(m,2\tau)} - \Lambda_{\boldsymbol{\theta}}^{(m,2\tau+1)} \end{pmatrix}}_{\Lambda_{\boldsymbol{\theta}}^{(m+1,2\tau)}}.\end{aligned}$$

Noting that $\mathsf{C}_{\boldsymbol{\theta}}^{(m+1)} = \mathsf{C}_{\boldsymbol{\theta}}^{(m+1,0)}$ completes the proof.

The theoretical results of this chapter can be summarized as follows:

**Theorem 1** *Any symmetric, positive definite, digitally shift-invariant kernel of the form* (9) *scaled to satisfy* (4d)*, when matched with digital sequence data sites, satisfies assumptions (4a, 4b, 4c, 4d). The* fast Walsh-Hadamard transform *performs the fast Bayesian transform,* $\boldsymbol{b} \mapsto \mathsf{V}^H\boldsymbol{b}$*, in* $\mathcal{O}(n \log n)$ *operations.*

***Proof.*** Lemma 2 establishes that covariance kernels of the form (8) matched with shifted digital net data sites satisfy the fast Bayesian cubature assumptions

- (4a), since $\mathsf{V}$ is the Walsh-Hadamard matrix,
- (4b), since the first column and row of the Walsh-Hadamard matrix consist of all ones, and
- (4d), since all Walsh functions but the zeroth integrate to zero.

What remains to be shown is (4c), namely how $\mathsf{V}^H\boldsymbol{b} = \mathsf{H}\boldsymbol{b}$ can be calculated in $\mathcal{O}(n \log n)$ operations for arbitrary $\boldsymbol{b}$.

The computation of $\widetilde{\boldsymbol{b}}^{(m)} = \mathsf{H}^{(m)}\boldsymbol{b}^{(m)}$ is done iteratively in what we show to be $m2^m$ operations, where $\mathsf{H}^{(m)}$ is $2^m \times 2^m$. The proof proceeds by induction. For the case $m = 0$, the Hadamard matrix $\mathsf{H}^{(0)}$ is the scalar 1, and $\widetilde{\boldsymbol{b}}^{(0)} = \mathsf{H}^{(0)}\boldsymbol{b}^{(0)} = \boldsymbol{b}^{(0)}$, which requires no arithmetic operations. Now, assume that $\mathsf{H}^{(m)}\boldsymbol{b}^{(m)}$ requires $m2^m$ operations, and let $\boldsymbol{b}^{(m+1)} = (\boldsymbol{b}_U^{(m)T}, \boldsymbol{b}_L^{(m)T})^T$. Let $\widetilde{\boldsymbol{b}}^{(m+1)} = \mathsf{H}^{(m+1)}\boldsymbol{b}^{(m+1)}$, $\widetilde{\boldsymbol{b}}_U^{(m)} = \mathsf{H}^{(m)}\boldsymbol{b}_U^{(m)}$, and $\widetilde{\boldsymbol{b}}_L^{(m)} = \mathsf{H}^{(m)}\boldsymbol{b}_L^{(m)}$. It follows from the definition of the Walsh-Hadamard matrix that

$$\begin{aligned}\widetilde{\boldsymbol{b}}^{(m+1)} &= \mathsf{H}^{(m+1)}\boldsymbol{b}^{(m+1)} = \begin{pmatrix} \mathsf{H}^{(m)} & \mathsf{H}^{(m)} \\ \mathsf{H}^{(m)} & -\mathsf{H}^{(m)} \end{pmatrix} \begin{pmatrix} \boldsymbol{b}_U^{(m)} \\ \boldsymbol{b}_L^{(m)} \end{pmatrix} \qquad \text{by (10)} \\ &= \begin{pmatrix} \mathsf{H}^{(m)}\boldsymbol{b}_U^{(m)} + \mathsf{H}^{(m)}\boldsymbol{b}_L^{(m)} \\ \mathsf{H}^{(m)}\boldsymbol{b}_U^{(m)} - \mathsf{H}^{(m)}\boldsymbol{b}_L^{(m)} \end{pmatrix} = \begin{pmatrix} \widetilde{\boldsymbol{b}}_U^{(m)} + \widetilde{\boldsymbol{b}}_L^{(m)} \\ \widetilde{\boldsymbol{b}}_U^{(m)} - \widetilde{\boldsymbol{b}}_L^{(m)} \end{pmatrix}.\end{aligned}$$

Thus, to compute $\widetilde{\boldsymbol{b}}^{(m+1)}$ requires two matrix multiplications by $\mathsf{H}^{(m)}$, at a cost of $m2^m$ each, plus an addition and a subtraction of vectors of length $2^m$, for a total cost of $2 \times m2^m + 2 \times 2^m = (m+1)2^{m+1}$, which is exactly what is hypothesized. Since $\widetilde{\boldsymbol{b}}^{(m)} = \mathsf{H}^{(m)}\boldsymbol{b}^{(m)}$ requires $m2^m = n\log_2(n)$ operations to compute, assumption (4c) is satisfied. This completes the proof of Theorem 1. □

## 4 Numerical Experiments

The Bayesian cubature algorithm described here, using the digitally shift invariant kernel in (8) with order $r = 1$ and the Bayesian lattice cubature algorithm described in [12] have been coded as `cubBayesNet_g` and `cubBayesLattice_g`, respectively, in GAIL [4]. We illustrate our algorithm for three common examples, which are also discussed in [12]. The first example evaluates a multivariate Gaussian probability, the second example is Keister's function [13], and the final example is pricing an Asian arithmetic mean option. For each example we are able to find the true integral value, $\mu$, either by analytic computation, or by running a numerical algorithm with a very small error tolerance.

The nodes used in `cubBayesNet_g` are the randomly scrambled and shifted Sobol' points supplied by MATLAB's Sobol' sequence generator. Four hundred different error tolerances, $\varepsilon$, are randomly chosen such that $\log(\varepsilon)$ has a uniform distribution. Although $\log(\varepsilon)$ is uniformly chosen, the times are stratified since the number of samples must be a power of 2. For each integral example, each $\varepsilon$, and each stopping criterion—empirical Bayes, full Bayes, and generalized cross-validation—we ran `cubBayesNet_g`. For each run, the execution time is plotted against $|\mu - \widehat{\mu}_n|/\varepsilon$. We expect $|\mu - \widehat{\mu}_n|/\varepsilon$ to be no greater than one, but hope that it is not too much smaller than one, which would indicate that the stopping criterion is too conserva-

tive. Throughout the experiments we use $\eta_1 = \cdots = \eta_d = \eta$. All the tests were run in MATLAB version 2019b running on Intel i7-7700HQ.

### 4.1 Multivariate Gaussian Probability

This integral is formulated as in [12] following the variable transformation introduced by Alan Genz [9]. The problem is originally three-dimensional but is transformed to a $d = 2$ dimensional problem. The simulation results are summarized in Figs. 2, 3, and 4. In all cases, `cubBayesNet_g` returns an approximation within the prescribed error tolerance. For $\varepsilon = 10^{-5}$ with the empirical Bayes stopping criterion, `cubBayesNet_g` takes about 3 s as shown in Fig. 2 whereas using a Matérn kernel requires 30 s to obtain the same accuracy as shown in [12]. This highlights the speed-up possible using fast Bayesian cubature.

The algorithm `cubBayesNet_g` uses MATLAB's fast Walsh transform. This is slower than MATLAB's fast Fourier transform, which is implemented in compiled code. This may help explain why `cubBayesLattice_g` is faster than `cubBayesNet_g` for this example. Also, `cubBayesLattice_g` uses higher order kernels whereas higher order digitally shift-invariant kernels are inappropriate for these examples. On average, `cubBayesLattice_g` uses $n \approx 16{,}000$ samples for $\varepsilon = 10^{-5}$, whereas `cubBayesNet_g` uses $n \approx 32{,}000$ samples. Amongst the three stopping criteria for `cubBayesNet_g`, GCV achieves an acceptable approximation faster than others and is also less conservative.

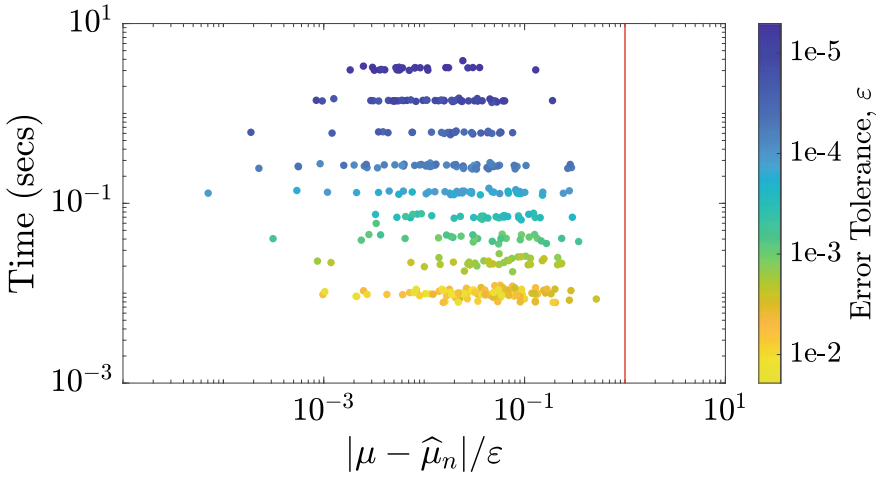



**Fig. 2** Multivariate normal probability example with empirical Bayes stopping criterion. Algorithm meets the error threshold for all the $\varepsilon$ randomly chosen in $[10^{-5}, 10^{-2}]$

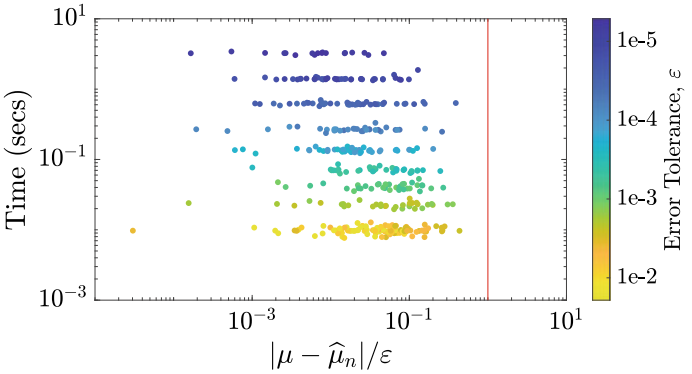



**Fig. 3** Multivariate normal probability example with the full-Bayes stopping criterion. Algorithm meets the error threshold for all the $\varepsilon$ randomly chosen in $[10^{-5}, 10^{-2}]$

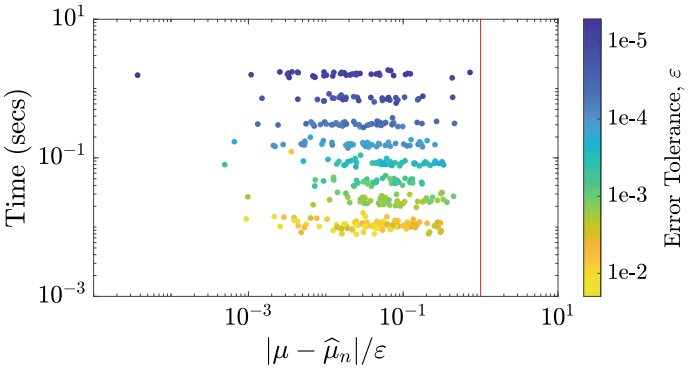



**Fig. 4** Multivariate normal probability example with the GCV stopping criterion. Algorithm meets the error threshold for all the $\varepsilon$ randomly chosen in $[10^{-5}, 10^{-2}]$

## *4.2 Keister's Example*

This multidimensional integral function comes from [13] and is inspired by a physics application:

$$\mu = \int_{\mathbb{R}^d} \cos(\|\boldsymbol{t}\|) \exp(-\|\boldsymbol{t}\|^2) \, \mathrm{d}\boldsymbol{t} = \int_{[0,1]^d} f_{\text{Keister}}(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x},$$

where

$$f_{\text{Keister}}(\boldsymbol{x}) = \pi^{d/2} \cos\left(\|\boldsymbol{\Phi}^{-1}(\boldsymbol{x})/2\|\right),$$

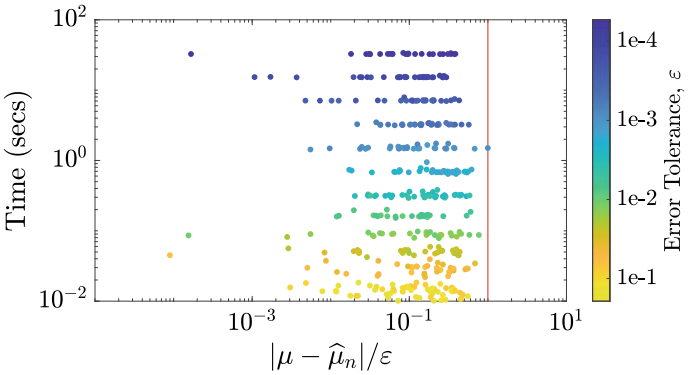



**Fig. 5** Keister example using the empirical Bayes stopping criterion. Algorithm meets the error threshold for all the $\varepsilon$ randomly chosen in $[10^{-4}, 10^{-1}]$

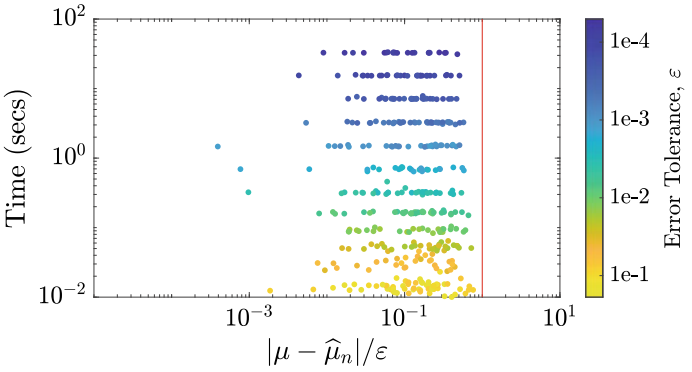



**Fig. 6** Keister example using the full-Bayes stopping criterion. Algorithm meets the error threshold for all the $\varepsilon$ randomly chosen in $[10^{-4}, 10^{-1}]$

and $\boldsymbol{\Phi}$ is the component-wise standard normal distribution. The true value of $\mu$ can be calculated iteratively in terms of quadrature as found in [12, Sect. 5.2].

Figures 5, 6 and 7 summarize the numerical tests for this case for dimension $d = 4$ and order $r = 1$. The `cubBayesLattice_g` algorithm in [12] uses a much smoother kernel than that used here. This explains why `cubBayesNet_g` uses more samples than `cubBayesLattice_g` on average. As observed from the figures, the GCV stopping criterion, in Fig. 7 achieves the results faster than the others but it is less conservative which is also the case with the multivariate Gaussian example.

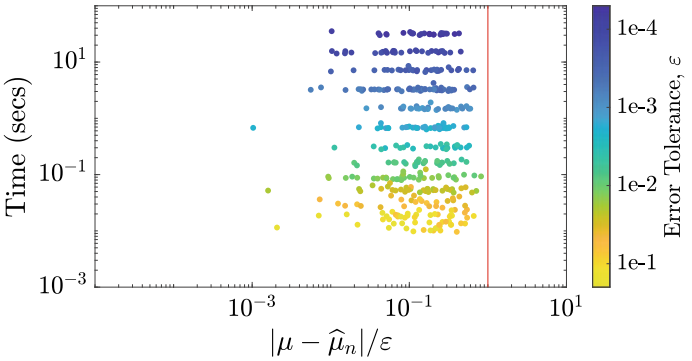



**Fig. 7** Keister example using the GCV stopping criterion. Algorithm meets the error threshold for all the $\varepsilon$ randomly chosen in $[10^{-4}, 10^{-1}]$

## *4.3 Asian Option Pricing*

The price of financial derivatives can often be modeled by high dimensional integrals. We refer to the formulation of the fair price of the option as in [12], where the underlying asset is described in terms of a discretized geometric Brownian motion.

Figures 8, 9 and 10 summarize the numerical results for the option pricing example using the values for time horizon $T = 1/4$ of a year, $d = 13$ time steps, initial asset price of $S_0 = 100$, interest rate of 0.05 per year, volatility of $\sigma = 0.5$ per root year, and strike price of $K = 200$, the same as used in the experiments with `cubBayesLattice_g` [12]. This integrand has a kink caused by the max function, so lattice rules cannot perform better, even if the integrand were to be periodized. We observe that `cubBayesNet_g` is more efficient in terms of the number

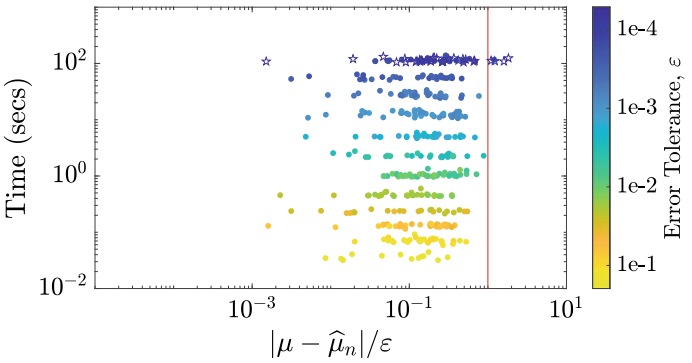



**Fig. 8** Option pricing using the empirical Bayes stopping criterion. The 33 hollow stars at the higher time indicate that the half-width of the credible interval did not meet the error threshold $\varepsilon$ before reaching maximum $n$

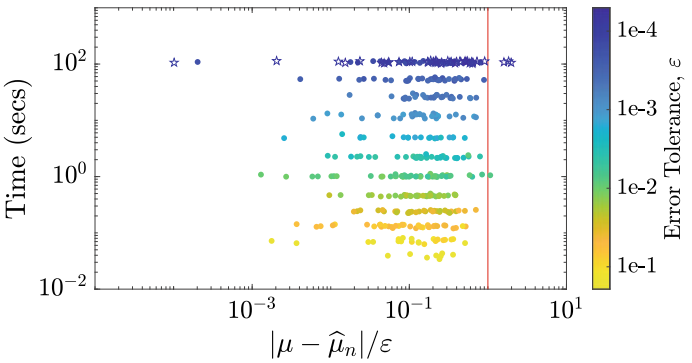



**Fig. 9** Option pricing using the full-Bayes stopping criterion. The 41 hollow stars indicate the half-width of the credible interval did not meet the error threshold $\varepsilon$ before reaching maximum $n$

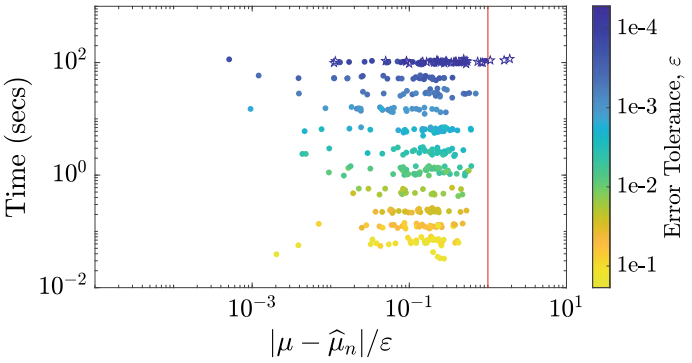



**Fig. 10** Option pricing using the GCV stopping criterion. The 35 hollow stars indicate the half-width of the credible interval did not meet the error threshold $\varepsilon$ before reaching maximum $n$

of samples used than `cubBayesLattice_g`. For the error tolerance, $\varepsilon = 10^{-3}$, `cubBayesLattice_g` uses $n \approx 2^{20}$ samples, whereas `cubBayesNet_g` uses $n \approx 2^{17}$ samples.

## *4.4 Discussion*

As shown in Figs. 2, 3, 4, 5, 6, 7, 8, 9 and 10, our algorithm computed the integral within user specified threshold with some exceptions. The exceptions occurred for the option pricing example due to the complexity and higher dimension of the integrand, the small tolerance, and the limit on the sample size. This indicates that fast Bayesian cubature may suffer in such cases and that further study is needed. Also notice that our algorithm `cubBayesNet_g`, finished within 40 s per run for the multivariate

Gaussian and Keister examples for the smallest tolerance as shown in Figs. 2, 3, 4, 5, 6 and 7. Option pricing took little above 100 s per run due to the complexity of integrand as shown in Figs. 8, 9 and 10.

A noticeable aspect from the plots of `cubBayesNet_g` is how close the error bounds are to the true error in many cases. This shows that the `cubBayesNet_g`'s error bounding is not too conservative.

In nearly all of the examples, the ratio $|\mu - \widehat{\mu}_n|/\varepsilon$ is closer to one for the Bayesian net cubature in contrast to the Bayesian lattice cubature. A possible reason is that the periodization transform speeds the convergence in the latter case and our data based error bounds are unable to capture this smaller error.

## 5 Conclusion and Future Work

We have extended our fast automatic Bayesian cubature to digital net sampling via digitally shift-invariant covariance kernels and fast Walsh transforms. Implementation of our algorithm `cubBayesNet_g`, is available in the Guaranteed Automatic Integration Library (GAIL) [4] and Quasi-Monte-Carlo Software in Python (QMCPy) [5]. We demonstrated `cubBayesNet_g` using three example integrands compared with `cubBayesLattice_g`. One major advantage of this algorithm, unlike the `cubBayesLattice_g` developed in [12], is that the integrand does not have to be periodic. However, unlike `cubBayesNet_g`, `cubBayesLattice_g` is more efficient for smoother, periodic functions when using sufficiently smooth and periodic covariance kernels.

The `cubBayesNet_g` in the current implementation uses only the first order kernel and digital nets. Accuracy and speed of the algorithm could be improved by using higher order digital nets and smoother digitally shift-invariant covariance kernels. This could help with the smoother integrands. This is a promising direction for future work.

For higher dimensions, both Bayesian cubature algorithms sometimes fail to produce an acceptable answer within a reasonable amount of time. This seems to be related to an excessive amount of time required to identify the optimal shape parameter $\boldsymbol{\theta}$. The root of the problem and its resolution is a matter for future investigation.

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